Operating Systems - CS(D)

Quiz-6 Solution

Q. Banker's Algorithm: Sequence of Execution and Deadlock Detection

Available Vector (1.5 marks)

Total resources are A: 12, B: 10, C: 12, D: 7. Sum the allocation matrix:

- A: $2+3+2+0+1+1+0=9$
- B: $1+2+1+1+2+1+1=9$
- C: $1+2+1+2+1+0+2=9$
- D: $0+1+2+0+1+1+1=6$

Available = Total - Allocated = $(12-9, 10-9, 12-9, 7-6) = (3,1,3,1)$.

Need Matrix (3.5 marks)

Need = Max - Allocation:

- P0: $(4-2, 2-1, 2-1, 2-0) = (2,1,1,2)$
- P1: $(5-3, 3-2, 3-2, 3-1) = (2,1,1,2)$
- P2: $(3-2, 2-1, 2-1, 3-2) = (1,1,1,1)$
- P3: $(2-0, 2-1, 3-2, 1-0) = (2,1,1,1)$
- P4: $(3-1, 3-2, 2-1, 2-1) = (2,1,1,1)$
- P5: $(4-1, 2-1, 1-0, 2-1) = (3,1,1,1)$
- P6: $(3-0, 3-1, 3-2, 2-1) = (3,2,1,1)$

Running Sequence (4 marks)

Start with Available: $(3,1,3,1)$. Find a sequence where each process's need is less than or equal to the available resources:

- P2 (Need: 1, 1, 1, 1) $\leq$ (3, 1, 3, 1), execute, release (2, 1, 1, 2), Available: (5, 2, 4, 3)
- P0 (Need: 2, 1, 1, 2) $\leq$ (5, 2, 4, 3), execute, release (2, 1, 1, 0), Available: (7, 3, 5, 3)
- P1 (Need: 2, 1, 1, 2) $\leq$ (7, 3, 5, 3), execute, release (3, 2, 2, 1), Available: (10, 5, 7, 4)
- P3 (Need: 2, 1, 1, 1) $\leq$ (10, 5, 7, 4), execute, release (0, 1, 2, 0), Available: (10, 6, 9, 4)
- P4 (Need: 2, 1, 1, 1) $\leq$ (10, 6, 9, 4), execute, release (1, 2, 1, 1), Available: (11, 8, 10, 5)
- P5 (Need: 3, 1, 1, 1) $\leq$ (11, 8, 10, 5), execute, release (1, 1, 0, 1), Available: (12, 9, 10, 6)
- P6 (Need: 3, 2, 1, 1) $\leq$ (12, 9, 10, 6), execute, release (0, 1, 2, 1), Available: (12, 10, 12, 7)

Safe sequence: **P2 → P0 → P1 → P3 → P4 → P5 → P6**.

Note: Any safe sequence that satisfies the Banker's Algorithm is acceptable for full marks in this step. Examples of other possible safe sequences include
P3 → P2 → P4 → P0 → P1 → P5 → P6,
P4 → P3 → P2 → P0 → P1 → P5 → P6,
P2 → P3 → P0 → P4 → P1 → P5 → P6, etc.

Final Verdict (1 mark)

Since a safe sequence exists, there is **no deadlock**.